

URLLC Performance Evaluation of IMT-2020 Candidate Technology: DECT-2020 New Radio

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Abstract—The European Telecommunications Standards Institute (ETSI) Technical Committee (TC) Digital Enhanced Cordless Telecommunications (DECT) and DECT forum have recently introduced DECT-2020 New Radio (NR) as an International Mobile Telecommunications 2020 (IMT-2020) candidate technology for the massive Machine Type Communications (mMTC) and Ultra Reliable Low Latency Communications (URLLC) use cases. To consider DECT-2020 NR as an IMT-2020 technology, the International Telecommunication Union, Radiocommunication Sector (ITU-R) has determined different independent evaluation groups to assess its performance against the IMT-2020 requirements. These independent evaluation groups are now in the process of investigating DECT-2020 NR. In order to successfully assess a technology, one important aspect is to fully understand the underlying physical layer and its performance in different environments. Therefore, in this paper, we first focus on the physical layer of DECT-2020 NR and investigate its link-level performance with standard channel models provided by the ITU-R for evaluation. We then use these link-level results to perform system-level evaluation of DECT-2020 NR for its suitability for the URLLC use case of Fifth Generation (5G). By performing extensive simulations, we show that DECT-2020 NR is well suited for URLLC. The results presented in this work will be beneficial for the independent evaluation groups and researchers as these results can help calibrating their physical layer performance curves and in comparing the system-level performance of DECT-2020 NR.

I. INTRODUCTION

Due to growing number of new applications day by day and their advanced requirements, the mobile communication industry witnesses a generation transition almost after every decade. To process this transition smoothly, different organizations regularly work together to create new standards, technical specifications, and regulations [1]. In the last years, the International Telecommunication Union, Radiocommunication Sector (ITU-R) Working Party 5D (WP5D) has set requirements for International Mobile Telecommunications 2020 (IMT-2020) to support Fifth Generation (5G) use cases involving Ultra Reliable Low Latency Communications (URLLC), massive Machine Type Communications (mMTC), and enhanced Mobile Broadband (eMBB) [2]. By targeting one or more of these 5G use cases, Third Generation Partnership Project (3GPP) has developed technologies such as New Radio (NR), Narrowband Internet of Things (NB-IoT), and Long Term Evolution for Machines (LTE-M) that meet the requirements set by ITU-R [3], [4].

One of the candidate technologies currently being evaluated, submitted by European Telecommunications Standards

Institute (ETSI) Technical Committee (TC) Digital Enhanced Cordless Telecommunications (DECT) and DECT Forum to ITU-R, is DECT-2020 NR [5]. DECT-2020 NR is a Radio Interface Technology (RIT) that mainly focuses on applications that fall between the URLLC and mMTC extremes, e.g., presence monitoring. DECT-2020 NR offers local deployment options without the need of a separate network infrastructure or network planning, and supports autonomous and automatic operation once deployed. Hence, minimal deployment or maintenance effort is required making it an attractive technology. The legacy DECT standard was originally designed to support cordless telephone systems in early 1990s, but is continually updated in response to the developments in the technology. Its latest development, DECT-2020 NR, targets local area wireless applications, and offers star and mesh network topology to support URLLC and mMTC use cases, respectively. The DECT-2020 NR is operated in frequency bands below 6 GHz, and despite its advanced physical layer numerology and medium access control algorithms, it can coexist with legacy DECT in current frequency bands allocated to legacy DECT.

The ETSI TC DECT has already provided detailed technical specifications of DECT-2020 NR [6]. However, to include DECT-2020 NR as an IMT-2020 technology, ITU-R has determined different independent evaluation groups to assess its performance against the IMT-2020 requirements. These independent evaluation groups are now in process of investigating DECT-2020 NR.

In order to successfully assess a technology, one important aspect is to fully understand the underlying physical layer and its performance in different environments. Therefore, in this paper, we first explain DECT-2020 NR in detail and then analyze the link-level performance under standard channel models. By using the presented link-level results, we then perform system-level simulations and evaluate the suitability of DECT-2020 NR for URLLC. We have been already collaborating with the proponents for the self-evaluation of DECT-2020 NR, and we believe that the results presented in this paper will be beneficial to independent evaluation groups as well as other researchers. These results will help understanding the physical layer of DECT-2020 NR better, in calibrating performance curves, and in analyzing system-level performance of DECT-2020 NR by either directly considering our physical layer results in new evaluations or comparing system-level results.

Our main contributions can be summarized as follows:

- We present a comprehensive overview of the newly introduced DECT-2020 NR standard (Section III).
- We implement its complete physical layer and develop the standard channel models provided by ITU-R (Section IV).
- Finally, we perform extensive simulations to evaluate the link- and system-level performance of DECT-2020 NR for the URLLC use case.

II. BACKGROUND AND RELATED WORK

DECT-2020 NR is a relatively new technology and, hence, does not have a vast literature. Details of DECT-2020 NR are provided by ETSI TC DECT in [5], whereas an overview is presented in [7]. The main focus of [7] is a system-level evaluation of DECT-2020 NR for the mMTC use case. Recently, an independent evaluation group has also assessed this technology and provided the insights briefly in [8]. Due to lack of some details, authors were unable to conclude whether DECT-2020 NR meets the requirement specified to become an IMT-2020 technology.

To be considered as an IMT-2020 technology, DECT-2020 NR must, amongst others, fulfil the requirement of 99.999 % reliability, user plane latency of less than 1 ms, and capability of supporting more than one million devices per km² [2]. These requirements are set by ITU-R WP5D to support different 5G use cases. The ITU-R is a regulatory body and its responsibility is to ensure efficient and interference free operations of different radio communication systems, whereas WP5D works within the ITU-R and oversees IMT systems (i.e., 3G onwards) in particular [1]. Both 3GPP and ETSI are standards developing organizations and are part of ITU-R WP5D. The 3GPP has been actively involved in developing technical specifications for mobile communication for more than two decades and it has developed technologies such as NR, NB-IoT, and LTE-M to be the part of IMT-2020 [3], [4]. Each of these technologies usually targets applications that are bounded to a particular use case. However, there are some applications that fall between the URLLC and mMTC extremes, e.g., remote light control and presence monitoring. The reliability and delay requirements in these applications are much stricter than the ones described by mMTC but are more relaxed in comparison to URLLC. Addressing such applications, ETSI TC DECT has recently introduced DECT-2020 NR as a candidate technology for IMT-2020.

DECT-2020 NR offers a completely new physical layer numerology, advanced medium access control algorithms, and supports star as well as mesh network topology [5], [6]. To be considered as an IMT-2020 technology, its performance needs to be evaluated against the requirements defined in IMT-2020. In the literature, link-level performance of new technologies is usually evaluated by using standard channel models [9]. The ITU-R has already provided different channel models to evaluate the IMT-2020 candidate technologies in [10]. Therefore, in this work, we use these channel models to evaluate the performance of DECT-2020 NR. The next section provides an overview of the DECT-2020 NR technology.

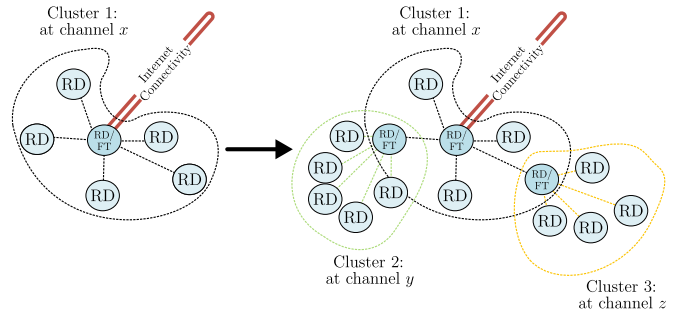


Figure 1: Formation of the mesh network topology.

III. DECT-2020 NEW RADIO

According to DECT-2020 NR technical specifications [5], any Radio Device (RD) in the network that has the capability of transmission and reception can be operated in either Fixed Termination (FT) mode or Portable Termination (PT) mode or both simultaneously. The RDs are free to choose any operation mode depending upon the local requirement. A RD operating in FT mode coordinates local resources, and provides information to other RDs on how to initiate a connection and communicate with it. The RD in FT mode is also responsible for routing the data either directly from the connected RDs to an external internet connection or through another RD in FT mode that has the access to the external internet connection. The RD in PT mode simply connects to the RD in FT mode for an indirect association to the external internet connection.

A star topology network supports URLLC use cases and is formed by one RD in FT mode while other RDs directly connect to it in PT mode. Whereas, a mesh network topology is realized by adding more RDs that connect themselves to any of the network nodes, hence, offering a scalable solution for mMTC use case as shown in Figure 1. In the mesh network topology, all RDs are capable of routing data though they autonomously choose or change their role between routing and non-routing depending upon the local decisions. Also, no central coordinator is needed and routing is based on a cost value rather than maintaining a routing table in each RD resulting in autonomous routing. Moreover, there can be unrestricted number of RDs connected to any type or number of external internet connections in a single network offering more than one route for other RDs to choose, hence, minimizing the probability of outage.

In general, DECT-2020 NR is targeted for frequency bands below 6 GHz. The physical layer utilizes Cyclic Prefix (CP) Orthogonal Frequency-Division Multiplexing (OFDM) in combination with Time Division Multiple Access (TDMA) as well as Frequency Division Multiple Access (FDMA) in a Time Division Duplex (TDD) mode for communication [6]. The OFDM subcarrier spacing is chosen between 27 kHz, 54 kHz, 108 kHz or 216 kHz depending on the subcarrier scaling factor $\mu = 1, 2, 4$ or 8, respectively. The size of the fourier transform depends on the scaling factor $\beta = 1, 2, 4, 8, 12$ or 16 which corresponds to Fast Fourier Transform (FFT) sizes of 64, 128, 256, 512, 768 or 1024 subcarriers.

The configuration $(\mu, \beta) = (1, 1)$ has the minimum nominal Radio Frequency (RF) bandwidth of 1.728 MHz while configuration $(\mu, \beta) = (8, 16)$ has the maximum nominal RF bandwidth of 221.184 MHz. After the FFT, 12.5% of the symbol are prepended as CP.

A single radio frame in DECT-2020 NR has a total duration of 10 ms and consists of 24 slots resulting in a slot duration of 416.67 μ s. This time segmentation was adopted from legacy DECT. A single uplink or downlink transmission must be a multiple of one slot, but not more than 16 slots in total. To allow even shorter transmissions, the slots are further divided into subslots. Each slot can accommodate 2, 4, 8 or 16 subslots depending upon the value of μ . Accordingly, a single transmission can also be a multiple of a subslot, but again may not exceed 16 subslots in total. It is noted that due to this partitioning, a subslot is always five OFDM symbols long, which is also the smallest possible transmission length.

Furthermore, on physical layer, the packet structure involves Synchronization Training Field (STF) and Data Field (DF), and is transmitted followed by a Guard Interval (GI) to avoid overlapping of transmissions in consecutive time slots. The STF is composed of training data for time and frequency synchronization at the receiving side, whereas the DF incorporates Physical Control Channel (PCC), Physical Data Channel (PDC), and Demodulation Reference Signal (DRS).

The physical layer in DECT-2020 NR is mainly responsible for modulation and demodulation, error detection and correction, Hybrid Automatic Repeat Request (HARQ) soft combining, signal synchronization, and for providing data to upper layers. Moreover, the physical layer also performs Multiple Input Multiple Output (MIMO) antenna processing and offers a possibility of realizing transmit diversity and beamforming. Error detection is performed by using a 16 or 24 bit Cyclic Redundancy Check (CRC), whereas for error correction, Turbo Coding is employed. Finally, the supported modulations include Binary Phase-Shift Keying (BPSK), Quadrature Phase-Shift Keying (QPSK), as well as different Quadrature Amplitude Modulation (QAM), i.e., 16-QAM, 64-QAM, 256-QAM, and 1024-QAM.

IV. SYSTEM MODEL

Given that DECT-2020 NR is an IMT-2020 candidate technology for 5G, we focus our evaluation on the requirements of the URLLC use case of 5G. These requirements are defined by the ITU-R in [2]. The most important requirements are a user plane latency below 1 ms and transmission reliability of at least 99.999 %. Both applies to layer 2 Protocol Data Unit (PDU) transmissions with a Transport Block Size (TBS) of at least 32 bytes. Furthermore, the ITU-R also specifies how the suitability of candidate technologies shall be evaluated to determine whether the requirements are met [10]. The evaluation procedure is divided into link-level and system-level simulations, which we will describe individually below. Furthermore, we describe in Section IV-B how the results are combined.

TABLE I: DECT-2020 NR Packet Formats for URLLC

Packet Property	Format 1	Format 2
(μ, β)	(4,1)	(4,2)
OFDM symbols	10	10
Transport Block Size (bit)	368	288
Modulation	QPSK	BPSK
Code rate	3/4	1/2
Bandwidth (MHz)	6.912	13.824
Subcarriers	64	128
HARQ retransmission	1	1
Transmit antennas	1 or 2 (TX diversity)	

A. Link-Level Details

Our goal on the link-level is to simulate Packet Error Rates (PERs) over Signal-to-Noise Ratio (SNR) to determine what the minimum SNR for 99.999 % reliability is. The first step towards this is to select suitable DECT-2020 NR packet configurations that contain at least 32 bytes and allow data transmissions within 1 ms. In this work, we focus on the two packet formats listed in Table I. We chose these formats because, on the one hand, they each allow one HARQ retransmission within 1 ms and, on the other hand, they utilize the two most secure modulations, i.e., BPSK and QPSK, respectively. The scaling factor $\mu = 4$ results from the requirement to have at least one HARQ retransmission, since for $\mu = \{1, 2\}$ the subslots in the time domain are too long. For Format 2 and BPSK, we utilize the factor $\beta = 2$ to increase the amount of data while maintaining the duration of a subslot. We assume a slotted data transmission based on subslots having a length of 52.08 μ s ($\mu = 4$). Two subslots (10 OFDM symbols) for data transmission are followed by one subslot for (negative) acknowledgement. Within the 1 ms user plane latency, a transmit and receive processing time of one subslot is implied.

After selecting the packet formats, the channel models for the link-level simulations must be selected. The ITU-R provides various channel models for Line-of-Sight (LOS) and Non Line-of-Sight (NLOS) scenarios [10], each consisting of path delays and corresponding average path gains, also known as Power Delay Profiles (PDPs). The channel parameters used in our simulations are summarized in Table II. The maximum Doppler spread of each channel can be inferred from a given carrier frequency and the maximum velocity of the User Equipments (UEs). The Doppler frequencies are then distributed according to the Jake's spectrum. Thus, the combination of a PDP and a Doppler Spread (DS) results in a doubly selective channel. Because the time distances between individual taps of the provided PDPs in [10] are partly shorter than one sample, all OFDM signals are oversampled to 27.648 MHz regardless of the net bandwidth. To generate the channel coefficients, we utilize the MATLAB MIMOchannel object, which we feed with the values from Table II.

Furthermore, we have to define the DECT-2020 NR receiver design whose most important parameters are listed in Table III. According to [10], the allowed maximum number or receive

TABLE II: Channel Parameters

Channel Property	Non Line-of-Sight	Line-of-Sight
Power Delay Profile Label [10]	TDL-iii	TDL-v
RMS Delay Spread	363 ns	93 ns
K factor	-	9 dB
Carrier Frequency	4 GHz and 700 MHz	
UE Velocity	3 km/h and 30 km/h	
Doppler Spread	1.9 Hz to 111.2 Hz	
Simulation Bandwidth	27.648 MHz	
Simulation Tool	MATLAB MIMOchannel	

TABLE III: Receiver Configuration

Receiver Property	Receiver configuration
Number of Antennas	8 (4 GHz) or 4 (700 MHz)
Diversity Combining	Maximum Ratio Combining
Time Synchronization	on first channel tap
Frequency Synchronization	ideal
Channel Estimation	Wiener filter
Channel Encoder/Decoder	MATLAB LTE Toolbox

antennas depends on the carrier frequency used. For 4 GHz, each UE can have up to 8 antennas, whereas for 700 MHz the number is limited to 4. For our link-level simulations, we consider only the maximum number of receive antennas, i.e., either 4 or 8, which are used for diversity combining by means of Maximum Ratio Combining (MRC).

For packet synchronization, we assume that packets are received at the first channel tap. This is a reasonable assumption as DECT-2020 NR exploits a slot-based system, therefore, time synchronization has a minor impact on the performance. For channel estimation, we utilize a two-dimensional Wiener filter. The channel coefficients are initially calculated by using an exponential PDP with the largest conceivable delay spread (i.e., 500 ns) and Jake's spectrum with the largest conceivable Doppler spread (i.e., 200 Hz). It is important to note that during operation, the Wiener filter coefficients remain the same, even if the instantaneous channel statistics deviate from the aforementioned assumptions [11]. Thus, our Wiener filter is a realistic design capable of being implemented in hardware.

For channel decoding we use a Turbo Decoder. Due to the close affinity of the channel coding of Long Term Evolution (LTE) and DECT-2020 NR, existing solutions can be used. The main differences concern the size of the transport blocks, the memory size of the soft buffer for HARQ processing, and the scrambling sequences. For our receiver, we use the channel coding from the LTE toolbox in MATLAB and incorporate it into our model in compliance with the DECT-2020 NR standard.

B. System-Level Details

For the link-level evaluation, we exploit formats that meet the low latency requirement. Furthermore, we can observe from the link-level results at what minimum SNR the desired PER of $1e-5$ will be achieved. However, a system-level evaluation is required to determine if that SNR will

TABLE IV: System-Level Parameters

Property	Value
	Configuration A B
Carrier frequency for evaluation	4 GHz 700 MHz
BS antenna height	25 m
Total transmit power per BS	36 dBm/MHz
UE power class	23 dBm
Percentage of high/low loss building	100% low loss
Inter-site distance	500 m
BS antennas	Sector Antenna (3 dB beamwidth of 65°)
Number of UE antenna elements	Up to 8 4 TX/RX
Device Deployment	80% outdoor, 20% indoor
UE mobility model	Fixed and identical speed $ v $ of all UEs, randomly and uniformly distributed direction
UE speeds of interest	3 km/h indoor 30 km/h outdoor
UE noise figure	7 dB
UE antenna element gain	0 dBi
Thermal Noise Level	-174 dBm/Hz
Traffic Model	Full Buffer
UE density	10 UEs per sector
UE antenna height	1.5 m

materialize at enough UEs in a realistic network scenario as the corresponding Signal-to-Interference-plus-Noise Ratio (SINR). More precisely, we need to determine the Cumulative Distribution Function (CDF) of all downlink or uplink SINRs and assess whether the SINR at the 5% percentile is above the SNRs required for a PER of $1e-5$. If this is the case, the technology also meets the requirement for reliability and thus for URLLC overall. This corresponds the URLLC evaluation procedure from [10].

The network scenario for our system-level simulation is described in [10]. It consists of base stations arranged in a hexagonal grid with an inter-site distance of 500 m. Each base station covers three sectors with an angular separation of 120° . Around the base stations, moving UEs are evenly distributed, each communicating to one base station through one of the sector antennas. All simulation parameters are listed in Table IV.

According to [10], we can simulate either downlink or uplink. In our case we consider downlink, i.e., the base stations transmit DECT-2020 NR signals to the UEs. Each UE is associated to the base station with maximum receive power. Due to UE movement, cell associations can change during simulation, however, we assume a seamless handover without additional delay. The UE movement is also incorporated at the link-level as an impairment due to Doppler shift. Neighbouring base stations can use different carrier frequencies. We consider three frequency reuse factors. For reuse factor 1, each base station uses the same frequency which results in the most interference from neighbouring base stations. For reuse factor

3 and 7, interference is reduced accordingly. Each UE can be indoor or outdoor. In case it is indoor, additional attenuation due to the walls of the building is included. Whether a UE has a LOS or NLOS channel is randomly assigned depending on the distance. Furthermore, correlated shadow fading is calculated for the entire network, which is incorporated in the path loss. All equations and parameters can also be found in [10].

All SINRs at the system level are single-port SINRs, the gains due to multi-antenna configurations are included at the link level. In addition to the SINR, we also measure the SNR, which represents interference-free reception and can be used as a comparative reference.

V. RESULTS AND DISCUSSION

To evaluate the URLLC performance of our system with the configurations presented in Section IV, we conducted Monte Carlo experiments on the link level to measure the PER at the receiver and on the system level to measure the CDF of the downlink SINR. Each link-level experiment ran until a convergence at a PER of $10e-5$ or less was obtained.

We first evaluate the link-level performance of the first ten Modulation and Coding Schemes (MCSs) from [6] over an Additive White Gaussian Noise (AWGN) channel, and plot the results in Figure 2. We provide these results as a comparative benchmark so that the performance of the channel decoder as a fundamental part of the receiver can be compared within the scope of DECT-2020 NR. The MCS 0 corresponds to BPSK with code rate $1/2$, whereas MCS 1 and MCS 2 represent QPSK with code rate $1/2$ and $2/3$, respectively. Similarly, MCS 7 corresponds to 64-QAM with code rate $5/6$ and so on. The size of the packet is fixed to $(\mu, \beta) = (1, 1)$ and a single slot is considered, which implies that the increasing modulation order also increases the TBS. As expected, for higher modulation order, high SNR value is required to achieve the same PER performance. These results are inline with the theory and the trend is similar to what we observe for LTE. However, these results cannot be directly compared with LTE due to marginally different TBSs here.

Next we consider link-level evaluation for URLLC. We consider in total six cases, which are depicted in Figure 3. Each of the six results shows four curves: the PER for the LOS channel with and without HARQ retransmission, and the NLOS channel, also with and without HARQ. By considering Configuration A (i.e., 4 GHz according to Table IV), the first two figures, Figure 3a and Figure 3b, show the PER for packet format 0 and packet format 1, respectively. We consider the maximum antenna diversity, i.e., two transmit antennas with transmit diversity and eight receiver antennas exploiting MRC. Due to the maximum antenna diversity, low PERs are observed for relatively low values of the SNR. For instance, in Figure 3b, the minimum SNR required for a PER of $10e-5$ is only -4 dB without HARQ retransmission and -7 dB with one HARQ retransmission. Moreover, it can be observed that with the considered antenna configuration, the performance

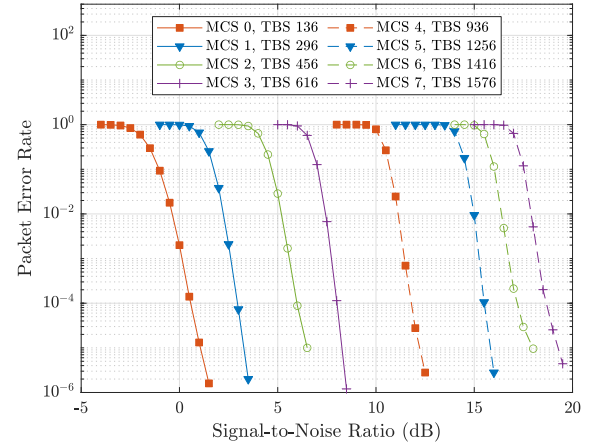


Figure 2: Packet error rate over different signal-to-noise ratio values in the AWGN channel under the assumption of perfect channel knowledge.

differences between the LOS and NLOS channels become marginal, accordingly the curves are very close.

The remaining four results in Figure 3 refer to Configuration B (i.e., 700 MHz according to Table IV). Since for this carrier frequency only four antennas are allowed at the receiver, we reduce the number of receive antennas accordingly. In Figure 3c and Figure 3d we consider two transmit antennas with transmit diversity and four receive antennas exploiting MRC, whereas in Figure 3e and Figure 3f we consider one transmit antenna and four receive antennas with MRC. The last two results have the lowest antenna diversity. Overall, we notice that except for Figure 3e, a $PER = 10e-5$ is achieved for $SNR \leq 1$ dB for all the considered configurations if HARQ is used. In the worst case, i.e., Figure 3e, we require an SNR of about 3 dB with one HARQ retransmission for the NLOS channel. These thresholds can now be used to compare to the 5% percentile of the system-level evaluation.

Figure 4 shows the system-level results. In Figure 4a, the CDF of the SINR across all UEs is shown for Configuration A. Similarly, the results for Configuration B are depicted in Figure 4b. With Configuration A and frequency reuse factor 7, a SNR of 2.95 dB is required at the 5% percentile. In other words, 95% of all UEs experience an SINR of at least 2.95 dB. As discussed above, however, for all results in Figure 3 (except Figure 3e), $SNR = 1$ dB is already sufficient to achieve $PER = 10e-5$. This means that these cases satisfy the conditions for URLLC. For configuration B in Figure 4b, on the other hand, the SINR at 5% percentile is 1.95 dB for reuse factor 7. This implies that the result from Figure 3e fails to meet the URLLC conditions as we achieve $PER = 10e-5$ only at $SNR = 3$ dB. However, it is not mandatory that all configurations satisfy the URLLC conditions. Moreover, many configurations with frequency reuse factor 3 are also satisfied, especially for the maximum antenna diversity, i.e., 2×8 MIMO with HARQ. Thus, our experiments show that DECT-2020 NR supports URLLC according to the ITU-R requirements.

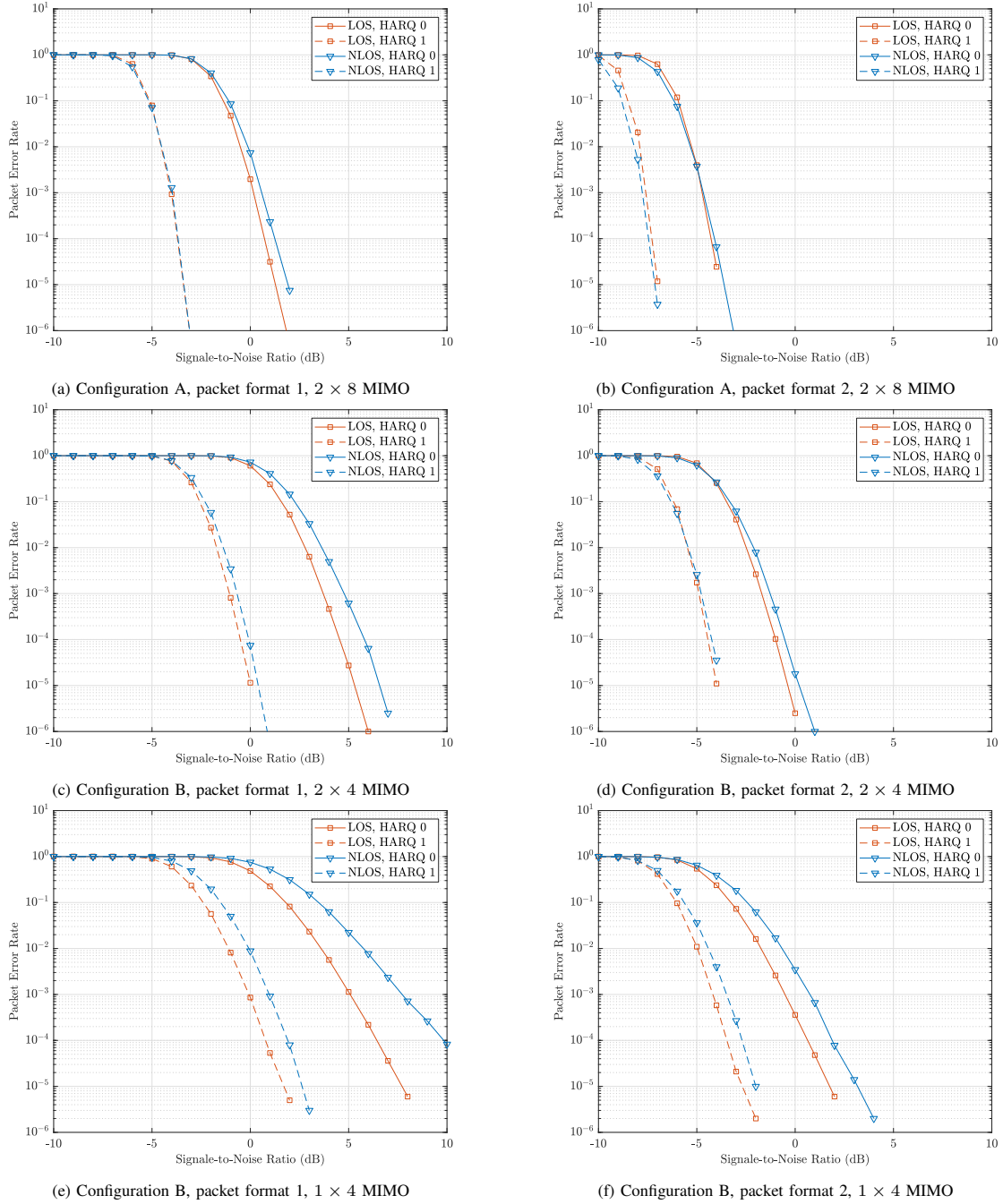
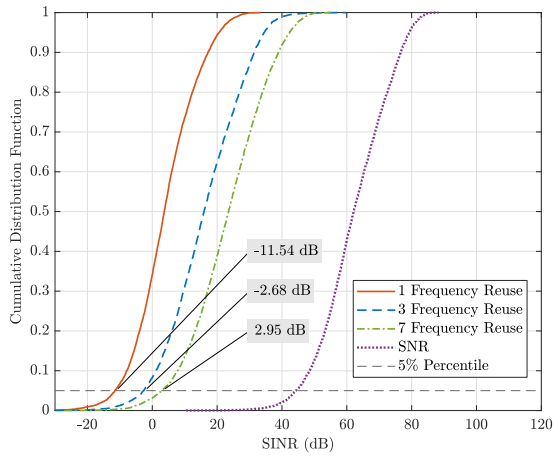


Figure 3: Packet error rate for Configuration A (i.e., 4 GHz) and Configuration B (i.e., 700 MHz) according to Table IV. Both packet format 0 and packet format 1 are considered for different MIMO configurations.

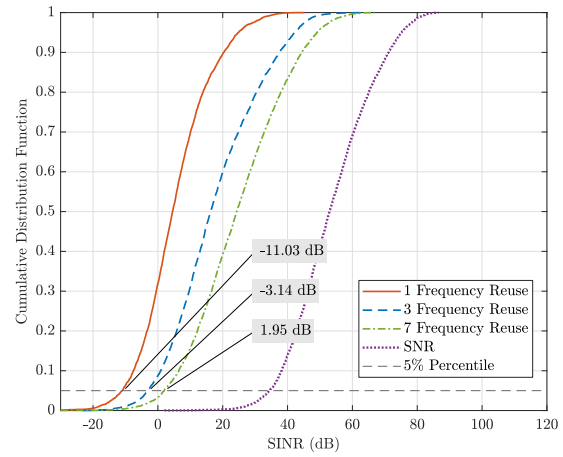
VI. CONCLUSION

This work evaluated the link- and system-level performance of the novel IMT-2020 candidate technology, DECT-2020 NR. We first provided an overview of DECT-2020 NR and presented possible packet configurations that meet latency and packet size requirements for the URLLC use case. We then considered the ITU-R evaluation procedures and parameters to evaluate the DECT-2020 NR performance and determine whether it meets the requirements set for the URLLC. Our

results show that DECT-2020 NR satisfies the conditions for URLLC according to ITU-R, i.e., the considered packet formats allow latencies below 1 ms with sufficient reliability of 99.999 % in realistic network scenarios with multiple base stations and UEs. These results were primarily obtained by using antenna diversity and HARQ. In the future, we plan to evaluate the DECT-2020 NR for the mMTC use case as well. Contrary to URLLC, the requirements of mMTC are lenient in terms of latency, instead the target is a large number of devices per unit area which make system-level



(a) Configuration A



(b) Configuration B

Figure 4: System-level results for Configuration A (i.e., 4 GHz) and Configuration B (i.e., 700 MHz) according to Table IV.

simulations more challenging. Furthermore, our goal is to develop a transceiver prototype, since there is no available solution yet.

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